

LESSON GUIDE

Technology Plus

Unit 5: Software

3.5.1 – Common Uses of AI

Lesson Overview:

In this lesson, students will explore how common tools like chatbots, assistants, and recommendation systems use artificial intelligence and what that means for how we interact with technology every day.

Students will:

- Identify common real-world uses of AI, including chatbots, assistants, and predictive systems .
- Compare how these systems behave and what they're designed to do.
- Explain how AI uses data to make decisions – and how that can go wrong.

Guiding Question: How does AI show up in the tools we already use – and what are its strengths and limitations?

Suggested Grade Levels: 8-12

Technology Needed: Yes

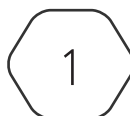
Approximate Time Required: 60-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.5 – Identify common uses of artificial intelligence (AI)
 - AI chatbots
 - AI assistants
 - Generative AI
 - AI-generated code
 - AI-generated content
 - AI predictions and suggestions

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Lesson Background

Background Information:

AI shows up in everyday tech more often than most people realize – from suggested playlists and auto-filled search bars to smart assistants and customer service bots. These tools don’t “think” like people do, but they do follow patterns, learn from data, and make decisions based on that training. Helping students recognize where AI is at work – and how it works under the hood – builds digital awareness and lays the foundation for more critical conversations about privacy, misinformation, and trust in future lessons.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.5.1 – Common Uses of AI • Activity Slides 3.5.1 – How Does AI Function
Materials per Group	• Computer with internet access for research
Materials per Student	• Student Notes 3.5.1 – Common Uses of AI • Student Handout 3.5.1 – Common Uses of AI • Assessment 3.5.1 – Common Uses of AI (optional) • Crayons, markers, or colored pencils

Lesson Background

Cyber Terms:

Term	Definition
artificial intelligence (AI)	a type of computer technology that can learn, adapt, and make decisions based on data
AI chatbot	a program that talks to users using text or voice, often to help with customer service or basic tasks
AI assistant	a more advanced AI tool that can complete tasks, answer questions, and control devices (like Siri or Alexa)
prediction AI	AI that guesses what might happen next using patterns in past data
suggestion AI	AI that recommends things based on your habits, searches, or past actions
Natural Language Processing (NLP)	a way for AI to understand and respond to human language (spoken or written)
machine learning	when AI adjusts its rules based on experience and improves over time by learning from data
personalization	when AI adjusts its responses or recommendations to fit your preferences or behavior
bias	when AI makes unfair decisions or suggestions because of flawed or incomplete training data
training data	the examples and information AI is given so it can learn how to make decisions

3.5.1 – Common Uses of AI

Guiding Question: How does AI show up in the tools we already use – and what are its strengths and limitations?

Lesson Launch (5-10 minutes)

- Start the lesson with a quick challenge. Say “Think about your day so far. What apps or tools have you already used today that might involve AI? Don’t worry about being right, just take 30 seconds to list anything that comes to mind.”
- After 30 seconds, direct students to partner up with and compare ideas.
- Allow 1-2 minutes for discussion and ask:
 - Did your partner think of something that you didn’t?
 - Did they think of something that surprised you?
 - Who thought of something that you weren’t sure was AI but maybe it is?
- Don’t overcorrect. Let students toss out guesses even if they are a little off.
- Examples to steer toward if needed:
 - Social media or YouTube recommending videos (Predictive AI)
 - Face ID or fingerprint unlock (Not AI - a biometric security feature that uses pattern matching)
 - Autocorrect/predictive text (Suggestive AI)
 - GPS rerouting (Not AI - more of an edge case that uses crowd-sourced data and may use machine learning)
 - Email spam filters (Predictive AI)
 - Smart speakers (AI assistants like Alexa, Siri)

The Real Intelligence on AI (15-20 minutes)

- Distribute Student Handout 3.5.1 – Common Uses of AI and encourage students to complete the handout as you go through today’s lesson.
- This section uses Slides 4-14 to introduce the foundational types of AI students already interact with - chatbots, assistants, and predictive/suggestive systems, while reinforcing how AI works and why it sometimes gets things wrong.
- Begin by distinguishing regular programs from AI systems. Emphasize that AI adjusts its rules based on experience (machine learning). Let students give examples - calculator vs. chatbot; static vs. adaptive behavior.
- Introduce AI chatbots. Use examples like “Track my order” or pop-up website chats. Explain the types

(rule-based, AI-powered, hybrid).

- Prompt: “Where have you interacted with a chatbot? Was it helpful?”
- Move into AI assistants. These are more powerful, multitasking tools like Alexa, Siri, etc. Show their versatility and why people use them.
- Prompt: “What do you use your AI assistant for? What does it do well – or poorly?”
- Cover predictive and suggestive AI. Use familiar apps (Netflix, Spotify, Amazon) to anchor the concepts. Clarify how AI makes predictions based on data patterns – not actual understanding.
- Wrap the section by showing limitations. Highlight that AI is pattern-based, not meaning-based. Common issues: bad training data, overconfidence, or bias.
- Share the tip: “Double-check. Be thoughtful about what you share.”
- Use slide 14 to pivot into Activity 1 by asking: “Now that we’ve seen what AI can do... how can we spot what it is doing in the real world?”

Activity 1: Bot or Not (10-15 minutes)

- In this activity, students analyze a variety of real-world tech scenarios and decide whether the technology described is not **AI**, a **chatbot**, an **assistant**, or a **predictive/suggestive AI** system. They’ll color-code each based on which type they believe it is - just one per box.
- **Have students select four different colored pencils, crayons, or markers** and assign one color to each category:
 - Not AI
 - Chatbot
 - Assistant
 - Prediction/Suggestion
- **Read the directions** aloud or paraphrase: “For each scenario, decide which category fits it best – then color the box the matching color. Only choose one per scenario, so pick the best fit.”
- Encourage students to **think critically** about the function of each tool. This isn’t just a recall activity, it’s about applying what they learned from the slides.
- After students finish, have them work in small groups to compare answers and **debate tricky ones**.
- Pick 2-3 examples for a whole class discussion: “Which category did you choose - and why?”
- Let students **justify their answers**, especially on the “tricky” scenarios. You’ll get great insight into how they’re thinking.
- If short on time, assign the Bot or Not Thinking Links activity as individual work and follow up with a quick check-in or quiz.
- For students who finish early: ask them to come up with **a fifth category** they think will show up in the

future (e.g., “AI that reads your emotions”) and make up their own sample scenario.

Activity 2: How Does AI Function? (20-25 minutes)

- This interactive simulation helps students explore how an AI system might "generate" an answer based on data from various sources, good, bad, or just plain weird. Students work in groups with assigned roles and discover how the quality of input affects the output.
- Use Activity Slides 3.5.1 – How Does AI Function to walk students through the directions.
- Students are placed into groups (4–5 students per group works well).
- Each group is assigned a **source type** like government websites, news articles, or social media posts – and must “answer” a question based solely on that source’s perspective.
- Remind your social media group that social media is more than just Snapchat and Instagram. When an AI pulls from social media, it could be Reddit, Quora, TikTok, Discord, or even YouTube comments - places where accuracy competes with attention. If your school blocks many of these sites, encourage students to write their answers in the style of posts or replies they might see on those platforms.
- After each group creates their source-based answer on Student Handout 3.5.1 – Common Uses of AI, the **AI Team** must combine **all the answers** into one unified “AI response.”
- To save time, consider letting the AI Team roam the room while other groups are working so they’re familiar with the different responses before finalizing their own.
- The AI group **cannot reject or fact-check** anything - they must treat all input as valid, just like a Large Language Model might do when trained on a messy internet.
- Have the AI group share their answer aloud so that other groups can record that answer onto their handout.
- The process will repeat in **Round 2**, but this time, instruct your Social Media group to lean into the wilder side of the internet. Their answers can be emotional, exaggerated, satirical, or even flat-out wrong. Tell them to “be cool” about it - don’t tip off the other groups. The goal is to introduce misinformation into the mix without the AI team realizing it until they must incorporate it into their final response.
- Once again, the AI group share their answer to this new prompt aloud and discuss what happened to their result. If the AI group’s Round 2 answer doesn’t seem affected by the misinformation, that’s okay humans have judgment, and they might ignore what feels unreliable. To highlight the contrast with how real AI systems work, ask the AI group what they noticed as they were collecting responses. Did anything stand out? Was there anything they chose not to use? This opens the door to discussing how AI doesn’t have the same filters we do.
- Use the prompts from the Student Handout to guide the reflection as you close this activity:
 - How did the AI group’s answers compare to the others?
 - What role did source quality play in shaping the final response?
 - Were any false or strange ideas repeated in the AI group’s summary? Why?
 - What would happen if someone trusted an AI trained on poor information?

Putting It All Together (5-10 minutes)

Discussion Questions

- **Say:** “That AI simulation gave us a taste of what it’s like to build an answer using lots of different sources, some better than others. But what does that mean for the apps and tools you use? Let’s bring it back to you.”
- **Ask:** “How can you tell if a tool or app you’re using is powered by AI?”
Student answers may vary but could include, “Think about the patterns it responds to, how it personalizes, or if it “learns” as you use it.”
- **Ask:** “Why is it important to know how AI works, even if you’re just using it?”
Student answers may vary but could include, “Understanding it helps you use it more effectively and avoid blindly trusting it.”
- **Ask:** “What risks come with using AI if you don’t understand where it gets its information?”
Student answers may vary but could include, “Think about bias, bad data, or misinformation in the AI’s training.”
- **Ask:** “What’s one thing you’ll do differently next time you interact with an AI system like a chatbot or assistant?”
Student answers may vary but could include, “Could be fact-checking, noticing behavior, being more aware of what data you’re giving it.”
- **Say:** “AI isn’t just something tech companies or scientists deal with, it’s something you already use every day. The more you understand how it works, the smarter you can be about how to use it, question it, and protect your data. Today you didn’t just learn what AI is, you learned how to spot it, evaluate it, and think critically about its role in your life.”